

Matrix Theory: Lecture 5

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Outline

- The Jordan Form (Nilpotents)
- Uniqueness of blocks
- The General Jordan Form



The Jordan Normal Form (for Nilpotents)

Theorem 7.5

For every nilpotent operator $\mathcal{A} : V \mapsto V$ there exists a basis of V consisting of s chains of length k_i , $i = 1, \dots, s$, with $\mathcal{A}^{k_i} v_i = 0$. In this basis the operator $\mathcal{A}$ has the matrix

$$J = \begin{pmatrix} J_{k_1}(0) & 0 & \dots & 0 \\ 0 & J_{k_2}(0) & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & J_{k_s}(0) \end{pmatrix},$$

where $J_k(0)$ is a Jordan block (of size $k \times k$ with eigenvalue 0) of the type

$$J_k(0) = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & 0 & \dots & 0 & 0 \end{pmatrix}$$

The Jordan Normal Form (for Nilpotents)

Theorem 7.6

For any nilpotent operator the number n_k of the blocks of size k is the same for any normal form. More exactly they can be found from the equations:

$$\begin{aligned}p_1 &= \dim \operatorname{Ker} \mathcal{A}^1 = n_1 + n_2 + n_3 + n_4 + n_5 + \cdots, \\p_2 &= \dim \operatorname{Ker} \mathcal{A}^2 = n_1 + 2n_2 + 2n_3 + 2n_4 + 2n_5 + \cdots, \\p_3 &= \dim \operatorname{Ker} \mathcal{A}^3 = n_1 + 2n_2 + 3n_3 + 3n_4 + 3n_5 + \cdots, \\p_4 &= \dim \operatorname{Ker} \mathcal{A}^4 = n_1 + 2n_2 + 3n_3 + 4n_4 + 4n_5 + \cdots, \\&\dots\dots\dots\end{aligned}$$

Note that evidently $n_k = 0$ if $k > \dim V$.



The General Jordan Normal Form

Def. For any $\lambda \in \mathbb{K}$ the **Jordan block** $J_k(\lambda)$ of size k looks like:

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{pmatrix}$$

thus λ on the main diagonal and $k - 1$ ones on the next. The rest is zero.
For size 1 we have only λ :

$$J_1(\lambda) = \begin{pmatrix} \lambda \end{pmatrix}.$$



The General Jordan Normal Form

Theorem 7.8

For every operator $\mathcal{A}$ there exists a basis consisting of chains of form

$$\begin{array}{ccccccc} (\mathcal{A} - \lambda_1 I)^{k_1-1} v_1, & (\mathcal{A} - \lambda_1 I)^{k_1-2} v_1, & \cdots, & (\mathcal{A} - \lambda_1 I) v_1, & v_1 \\ (\mathcal{A} - \lambda_2 I)^{k_2-1} v_2, & (\mathcal{A} - \lambda_2 I)^{k_2-2} v_2, & \cdots, & (\mathcal{A} - \lambda_2 I) v_2, & v_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{array}$$

with the conditions $(\mathcal{A} - \lambda_i I)^{k_i} v_i = 0$. All λ_i are eigenvalues and some of them may be equal. Relative to this basis the operator has matrix

$$J = \begin{pmatrix} J_{k_1}(\lambda_1) & O & \cdots & O \\ O & J_{k_2}(\lambda_2) & \cdots & O \\ \cdots & \cdots & \cdots & \cdots \\ O & O & \cdots & J_{k_m}(\lambda_m) \end{pmatrix},$$

which is called the **Jordan form** of the operator.

The General Jordan Normal Form

Theorem 7.9

For any eigenvalue λ of an operator, the number $n_k = n_k(\lambda)$ of blocks $J_k(\lambda)$ of size k is the same for any normal form. More exactly they can be found from the equations:

$$\begin{aligned} p_1 &= \dim \operatorname{Ker}(\mathcal{A} - \lambda I)^1 = n_1 + n_2 + n_3 + n_4 + n_5 + \cdots, \\ p_2 &= \dim \operatorname{Ker}(\mathcal{A} - \lambda I)^2 = n_1 + 2n_2 + 2n_3 + 2n_4 + 2n_5 + \cdots, \\ p_3 &= \dim \operatorname{Ker}(\mathcal{A} - \lambda I)^3 = n_1 + 2n_2 + 3n_3 + 3n_4 + 3n_5 + \cdots, \\ p_4 &= \dim \operatorname{Ker}(\mathcal{A} - \lambda I)^4 = n_1 + 2n_2 + 3n_3 + 4n_4 + 4n_5 + \cdots, \\ &\dots\dots\dots \end{aligned}$$

Thus the Jordan normal form is uniquely determined (up to the order of the blocks).



Consequences of the Jordan Form

Theorem 7.10

Two matrices are similar if and only if they have the same Jordan form.

Corollary 7.11

Every matrix A is similar to its transpose A^T .

Theorem 7.12

If $\lambda_1, \dots, \lambda_n$ are all roots of $p_A(\lambda)$ (calculated with multiplicity), that is

$$p_A(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n)$$

then $\det A = \lambda_1 \lambda_2 \dots \lambda_n$ and $\operatorname{tr} A = \lambda_1 + \lambda_2 + \dots + \lambda_n$. In particular for $n = 2$ we have

$$p_A(\lambda) = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A)$$